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AirBnb Data Analystics Project

Power BI

End-to-End Data Analytics & Business Intelligence

Project Focus
New York City Airbnb booking, revenue, pricing, occupancy, host, property, review, and neighborhood analytics using Microsoft Power BI.

The project demonstrates a complete analytics workflow covering data preparation, Power Query transformation, dimensional modeling, DAX measures, interactive reporting, and business insight generation.

Independent portfolio / tutorial project

1. Project Overview

The AirBnb Data Analystics Project is an end-to-end data analytics solution focused on understanding Airbnb activity across New York City. The project uses Power BI for data modeling,
DAX calculations, interactive visualization, and business reporting.

The simulated dataset contains more than **100,000 booking records** covering **January 2023 through December 2024**, with 80 property listings, 50 hosts, and 20 NYC neighborhoods.

2. Business Objectives

2.1 Primary Objectives

- Monitor revenue and booking performance across NYC neighborhoods.
- Analyze occupancy, RevPAR, average nightly rate, and review score.
- Identify patterns across hosts, property types, room types, and neighborhoods.
- Track monthly and yearly performance using MoM and YoY comparisons.
- Build an interactive Power BI solution that converts raw data into actionable business insights.

2.2 Technical Objectives

- Use Power Query for data ingestion, cleaning, and transformation.
- Apply Star Schema principles for Power BI data modeling.
- Develop reusable DAX measures using VAR, CALCULATE, and time-intelligence functions.
- Create a clean two-page Power BI report using native visuals.

3. Project Scope

• Five CSV source files: fact_bookings, dim_host, dim_property, dim_location, and dim_date.

• Power Query data profiling, cleaning, and transformation.

• Resolution of 10 documented data-quality issues.

• Star Schema consisting of one fact table and four dimension tables.

• 24 DAX measures organized into five display folders.

• Two-page interactive Power BI report using native visuals.

• Time-based analysis from January 2023 through December 2024.

4. Data Sources and Volume

File Name	Table Name	Rows	Columns	Description
fact_bookings.csv	fact_bookings	100,200	13	Central booking transaction table containing booking, revenue, review, pricing, and a
dim_host.csv	dim_host	50	6	Host profile information including name, tenure, response rate, and Superhost status
dim_property.csv	dim_property	80	11	Listing information including room type, property type, amenities, and base price.
dim_location.csv	dim_location	20	8	NYC neighborhood reference data including latitude, longitude, and ZIP code.
dim_date.csv	dim_date	1,461	11	Date dimension covering January 2023 through December 2024.

5. Data Quality and Transformation

The source dataset intentionally contains common data-quality issues. These are cleaned and standardized before analysis so that the resulting metrics are consistent and suitable for reporting.

#	Table / Column	Issue	Treatment
1	dim_host / host_name	3 null or empty host names.	Standardize as “Unknown Host”.
2	dim_host / host_response_rate	3 decimal values instead of percentages.	Standardize to percentage representation.
3	dim_property / bedrooms	5 null values for studio units.	Set studio bedroom count to 0.
4	dim_location / latitude	2 values contain extra spaces.	Trim and convert to numeric.
5	dim_date / month_name	Mixed month-name capitalization.	Standardize to title case.
6	fact_bookings / price	About 3,000 text-formatted currency values.	Remove currency formatting and convert to numeric.
7	fact_bookings / review_score	About 1,500 scores exceed 5.0.	Exclude invalid scores.
8	fact_bookings / minimum_nights	About 800 values are zero or negative.	Retain values of at least 1 night.
9	fact_bookings / availability_365	About 600 records contain invalid value 999.	Convert invalid values to null.
10	fact_bookings / booking_id	200 duplicate booking records.	Remove duplicate booking IDs.

Analytics principle: data-quality rules are applied before KPI calculation and visualization to improve consistency and reliability.

6. Data Model and Analytics Layer

6.1 Star Schema

The Power BI model follows a classic Star Schema with one central fact table and four dimension tables. Relationships use one-to-many cardinality with single-direction filtering from dimensions to the fact table.

Dimension	Key	Fact Table Key	Relationship
dim_date	date_key	fact_bookings[date_key]	1:*
dim_host	host_id	fact_bookings[host_id]	1:*
dim_property	listing_id	fact_bookings[listing_id]	1:*
dim_location	location_id	fact_bookings[location_id]	1:*

6.2 Modeling Standards

- Hide foreign-key columns in fact_bookings from Report View after relationships are established.
- Mark dim_date as the official Date Table using full_date.
- Use Import Storage Mode for dimension tables.
- Disable Power BI Auto date/time.
- Store all business measures in the dedicated ■ Measures table.
- Maintain clear dimension-to-fact filtering and avoid unnecessary bidirectional relationships.

7. DAX and Reporting Requirements

- Develop 24 DAX measures organized into five display folders.
- Use Title Case with spaces for measure names; avoid underscores.
- Use VAR for readable intermediate calculations.
- Use CALCULATE for controlled filter-context changes.
- Use time-intelligence functions for MoM, YoY, prior-year, and YTD analysis.
- Use reusable business measures rather than visual-specific calculations wherever practical.
- Use native Power BI visuals and interactive slicers for report navigation.

Technology Stack: Power Query · DAX · Microsoft Power BI

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Business Intelligence & Data Analysis Report

Revenue • Demand • Pricing • Occupancy • Hosts • Property & Room Segmentation

$176.08M	$235.56	48.8%	3.92	20.0%
Total Revenue	Average Nightly Rate	Occupancy Rate	Average Review Score	Cancellation Rate

Project purpose. This report translates the Power BI Airbnb analysis into business questions, analytical observations and decision-support use cases. The report is based on the dashboard screenshots and visible values supplied for the project.

Important note. The analysis is an independent portfolio/reporting exercise and does not represent an official Airbnb publication or endorsement.

Executive message: The dashboard connects financial performance with demand, pricing, utilization, customer experience and host/property segmentation. Its main value is the ability to move from what happened? to where should management investigate further?

1. Executive Overview

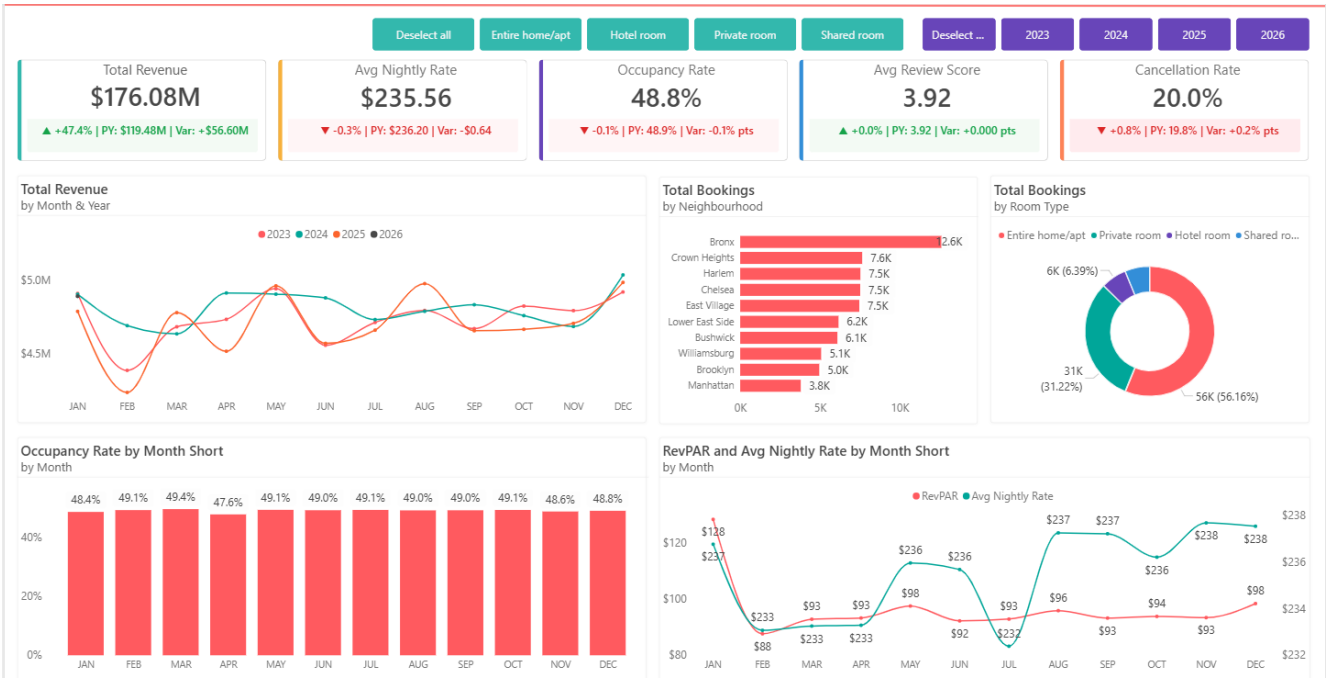
A management-level view of the business signals visible in the report.

Area	Observed signal	Business interpretation / decision use
Revenue	$176.08M; +47.4% vs PY	Strong reported revenue growth; investigate the operational drivers rather than attributing growth to price alone.
Pricing	$235.56; -0.3% vs PY	Average nightly pricing is broadly stable, so pricing alone does not explain the reported revenue increase.
Occupancy	48.8%; -0.1 pts vs PY	Utilization is broadly stable; investigate booking volume, inventory and mix for additional revenue drivers.
Customer experience	Review score 3.92	Average score is stable versus PY; useful for monitoring quality while revenue changes.
Cancellations	20.0%; +0.2 pts vs PY	A meaningful operational metric for lost/unstable demand and forecasting uncertainty.
Demand geography	Bronx highest in displayed ranking at 12.6K bookings	Booking demand is geographically uneven; location-level pricing, supply and marketing can be investigated.
Accommodation mix	Entire home/apt ≈56.16% of bookings	Whole-home inventory represents the largest visible room-type segment and merits separate pricing/revenue analysis.

Core analytical observation: Revenue is reported as substantially higher than the previous year, while average nightly rate and occupancy are both approximately stable. This means the dashboard should be used to investigate booking volume, inventory changes and segment mix rather than assuming that higher prices are the sole reason for the revenue movement.

2. Power BI Report — Page 1

Executive dashboard: KPIs, trends, neighborhood demand, room mix, occupancy and RevPAR.



Page 1 purpose: Provide a fast executive read on overall financial and operating performance, then give enough trend and segmentation context to identify areas for deeper investigation.

Interactive controls: The year buttons and room-type buttons allow the user to change the analytical slice without creating separate reports.

3. KPI Analysis — Business Logic and Decisions

Total Revenue — $176.08M

Business logic: Measures the aggregate revenue represented in the selected dashboard context. The displayed comparison is +47.4% versus PY, with a +$56.60M variance.

Decision use: Use it as the primary financial health indicator. The important follow-up is to decompose revenue into bookings, rate, occupancy and mix.

Average Nightly Rate — $235.56

Business logic: Measures the average amount paid per occupied night/listing in the dashboard's metric definition. The visible PY comparison is $236.20, a -$0.64 variance.

Decision use: Supports pricing strategy, market/segment comparison and evaluation of whether revenue growth is price-led.

Occupancy Rate — 48.8%

Business logic: Measures utilization of available accommodation capacity. The visible PY value is 48.9%.

Decision use: Useful for identifying underutilization and for evaluating whether pricing or demand-generation actions may be needed.

Average Review Score — 3.92

Business logic: Represents average customer review performance. The visible PY value is also 3.92.

Decision use: Provides a customer-experience quality signal that can be analyzed alongside bookings, revenue and property/host characteristics.

Cancellation Rate — 20.0%

Business logic: Measures the share of bookings that are cancelled under the dashboard's metric definition. The visible PY value is 19.8%.

Decision use: Useful for revenue-loss investigation, booking stability and operational forecasting.

4. Page 1 Chart Analysis

Total Revenue by Month & Year

What it tells us: Compares monthly revenue across 2023–2026. It is a seasonality and year-over-year diagnostic.

Business benefit: Identify high/low demand periods, investigate unusual monthly deviations, plan promotions and support revenue forecasting.

Total Bookings by Neighborhood

What it tells us: Ranks neighborhoods by booking volume; the displayed range is approximately 12.6K at the top to 3.8K at the bottom.

Business benefit: Shows where booking demand is concentrated. Combine with revenue and nightly rate before making location-level supply or pricing decisions.

Total Bookings by Room Type

What it tells us: Shows the composition of bookings by accommodation type. Entire home/apt is approximately 56.16%, followed by private room at about 31.22%.

Business benefit: Supports inventory and product-mix decisions. Compare booking share with revenue share to identify high-value segments.

Occupancy Rate by Month

What it tells us: Shows monthly occupancy values clustered around the high-40% range; the visible low is 47.6% in April and the visible high is 49.4% in March.

Business benefit: Useful for spotting utilization differences by month. Because the range is relatively narrow, occupancy alone does not explain the larger revenue movements.

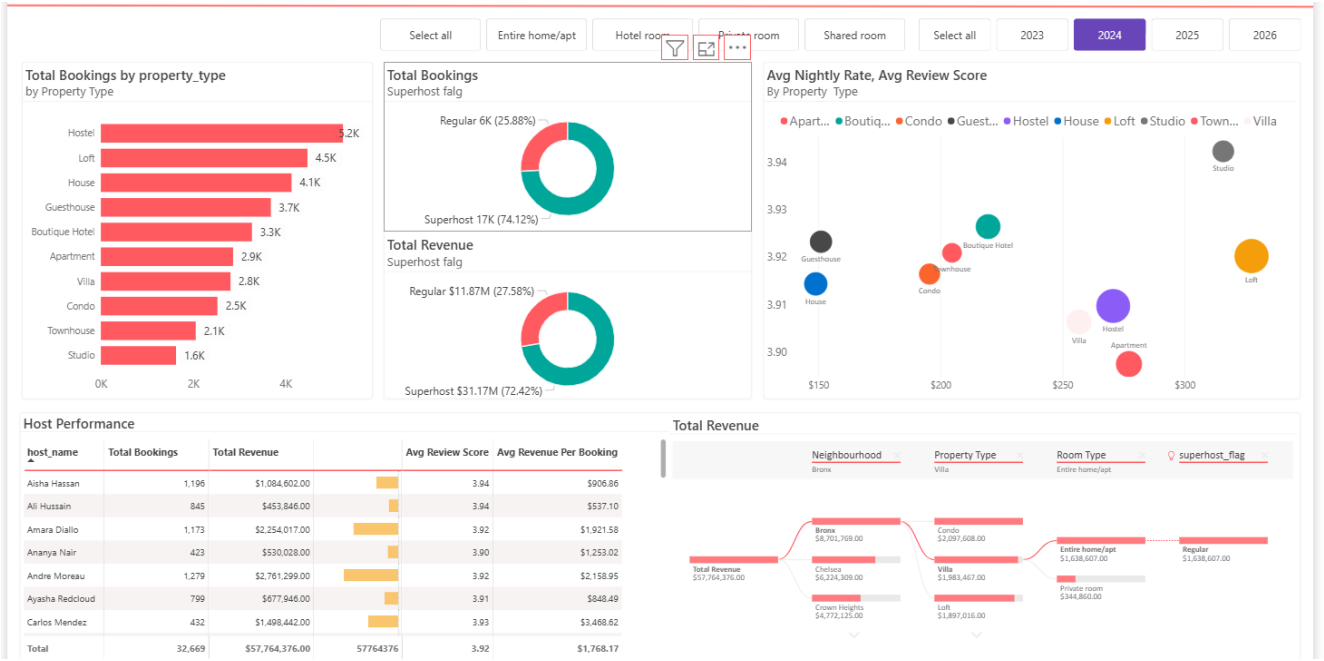
RevPAR and Average Nightly Rate by Month

What it tells us: Compares revenue per available room/unit with average nightly rate. RevPAR combines rate with utilization and is therefore a stronger efficiency measure than rate alone.

Business benefit: Helps determine whether revenue performance is more closely associated with pricing or utilization. High rate with weak RevPAR can signal underutilized capacity.

5. Power BI Report — Page 2

Detailed segmentation: property types, host status, pricing/reviews, host performance and revenue decomposition.



Filter context: The supplied screenshot shows the year selector with 2024 selected. Therefore, values visible on this page should be interpreted as the filtered 2024 view unless otherwise specified.

6. Page 2 — Property, Host and Revenue Analysis

Total Bookings by Property Type

What it tells us: Shows booking volume by property format. In the displayed 2024 view, Hotel is around 5.2K bookings, followed by Loft at 4.5K and House at 4.1K.

Business benefit: Supports inventory planning, property acquisition and property-type marketing. Booking volume should be combined with rate and revenue to assess economic value.

Total Bookings — Superhost vs Regular

What it tells us: The displayed 2024 donut shows Superhost at approximately 17K bookings (74.12%) and Regular at about 6K (25.88%).

Business benefit: Helps segment booking activity by host status and identify whether host characteristics are associated with different levels of demand.

Total Revenue — Superhost vs Regular

What it tells us: The displayed 2024 donut shows Superhost at about $31.17M (72.42%) and Regular at about $11.87M (27.58%).

Business benefit: Moves the analysis from booking volume to monetary contribution. Comparing booking share and revenue share helps investigate revenue per booking.

Average Nightly Rate vs Average Review Score by Property Type

What it tells us: A scatter/bubble view compares pricing with customer ratings across property types. It identifies segments with different price–rating combinations.

Business benefit: Useful for identifying premium, value and potential value-for-money segments for further investigation. It is descriptive, not causal.

Host Performance Table

What it tells us: Combines host-level bookings, total revenue, average review score and average revenue per booking.

Business benefit: Provides a multi-dimensional view of host performance. It prevents the analyst from judging hosts only by booking volume.

Revenue Decomposition Tree

What it tells us: Allows total revenue to be broken down interactively through Neighborhood → Property Type → Room Type → Superhost status.

Business benefit: Useful for finding revenue concentration and investigating combinations of dimensions that contribute to the selected revenue context.

7. Host Performance — Why Volume Alone Is Not Enough

The host table demonstrates why a business dashboard should distinguish between volume and value. For example, the visible table shows Andre Moreau with 1,279 bookings and about $2.76M revenue, while Carlos Mendez has 432 bookings but about $1.50M revenue. Carlos therefore has fewer bookings but substantially higher revenue per booking.

Business implication: A host-ranking process based only on booking count could overlook high-value hosts. Average revenue per booking provides an additional efficiency/value lens.

Host	Bookings	Revenue	Review	Avg Revenue / Booking
Aisha Hassan	1,196	$1,084,602	3.94	$906.86
Ali Hussain	845	$453,846	3.94	$537.10
Amara Diallo	1,173	$2,254,017	3.92	$1,921.58
Ananya Nair	423	$530,028	3.90	$1,253.02
Andre Moreau	1,279	$2,761,299	3.92	$2,158.95
Ayasha Redcloud	799	$677,946	3.91	$848.49
Carlos Mendez	432	$1,498,442	3.93	$3,468.62

These values are transcribed from the visible screenshot and are used here as examples of the analytical logic; the complete underlying dataset may contain additional hosts.

8. Decision-Support Matrix

Business problem	Dashboard evidence to use	Follow-up analysis / decision area
Revenue is changing	Total Revenue + monthly revenue trend	Decompose by bookings, rate, occupancy and segment mix.
Demand is uneven	Bookings by neighborhood	Compare neighborhood revenue, rate, occupancy and inventory.
Pricing needs review	Avg Nightly Rate + RevPAR	Look for high-rate/low-utilization and lower-rate/high-utilization segments.
Inventory mix needs planning	Room Type + Property Type	Compare booking share, revenue share and occupancy by segment.
Host performance differs	Superhost donuts + Host Performance table	Compare booking volume, revenue, review score and revenue per booking.
Customer experience needs monitoring	Avg Review Score	Drill into property type, host and room type to identify rating variation.
Bookings are unstable	Cancellation Rate	Analyze cancellations by month, neighborhood, room type, property type and host.
Revenue concentration needs explanation	Decomposition Tree	Drill through neighborhood → property type → room type → host status.

Recommended decision workflow: Start with the KPI change → identify the affected month/segment → compare bookings, rate and occupancy → drill into neighborhood/property/room type → evaluate host contribution → check cancellations and reviews before taking an operational action.

9. Analytical Methodology & Interpretation Notes

• Dashboard scope: The report is based on the two supplied Power BI screenshots and the visible KPI/chart values.

• Time context: Page 1 displays multi-year trends. Page 2 visibly has 2024 selected, so its values are treated as a 2024 filtered view.

• Descriptive vs causal: The dashboard identifies relationships and concentrations. It does not by itself prove that one factor causes another.

• Revenue vs bookings: High booking volume does not necessarily mean high revenue because nightly rates and segment mix can differ.

• Rate vs RevPAR: Average nightly rate measures price, while RevPAR incorporates available capacity/utilization and is therefore useful for efficiency analysis.

• Superhost analysis: The Superhost charts show performance by host-status segment; they should not be interpreted as proof that Superhost status causes higher bookings or revenue.

• Occupancy definition: The exact denominator and business rules should be documented from the source model before using the metric for external reporting.

• Currency: The dashboard displays monetary values with a dollar sign. This report preserves that presentation and does not independently convert currencies.

Recommended next-level analyses

1. Revenue and bookings by neighborhood together. 2. Revenue share vs booking share by room/property type. 3. Cancellation rate by segment. 4. Revenue per booking by Superhost status. 5. Review score vs bookings/revenue. 6. Monthly revenue forecasting. 7. Host segmentation into high-volume, high-value and high-rating groups.

10. Final Business Summary

The report provides a connected view of Airbnb business performance. The first page establishes the financial and operational picture: revenue, pricing, occupancy, reviews, cancellations and monthly/segment trends. The second page explains how those outcomes vary by property type, host status, host, price and review score, and provides an interactive path for decomposing revenue.

The most important analytical observation visible in the supplied dashboard is that reported revenue is substantially higher than the previous-year comparison while average nightly rate and occupancy are broadly stable. This makes booking volume, inventory changes and segment mix important areas for further investigation.

For a Data Analyst portfolio, the strongest aspect of this dashboard is not simply the number of charts. It is the connection between metrics and business questions: financial performance → demand → pricing → utilization → customer experience → host/property segmentation → revenue drivers.

Prepared as an independent portfolio analysis
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